



Pagerank part 2

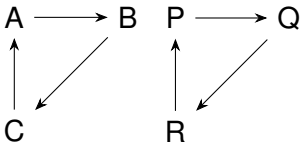
Hoa Vu

CS 663

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Naive PageRank's Flaws (Disconnected Components)

- Many solutions if the graph is disconnected.



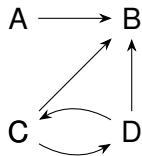
$$\pi_A = \pi_B = \pi_C = 1/3, \pi_P = \pi_Q = \pi_R = 0$$

$$\pi_A = \pi_B = \pi_C = \pi_P = \pi_Q = \pi_R = 1/6$$

...

Naive PageRank's Flaws (Dangling nodes)

- If a sink node exists, there might not be a solution.



$$\pi_A = 0$$

$$\pi_C = \pi_D/2$$

$$\pi_D = \pi_C/2$$

$$\implies \pi_C = \pi_D = 0$$

$$\implies \pi_B = 0$$

Dealing with Sinks

- Imagine an edge from each sink to every other node.

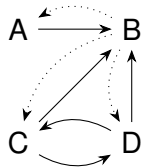


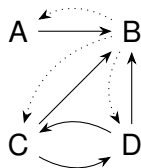
Figure: Now, $d_B = 3$.

Dealing with Disconnected Graphs

- Add a damping factor α .

$$\pi_v = \alpha \left(\sum_{u \rightarrow v \in E} \frac{\pi_u}{d_u} \right) + \frac{1 - \alpha}{n}$$

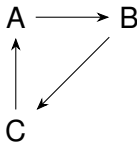
- Google set $\alpha \approx 0.85$.



$$\pi_c = \alpha (\pi_b/4 + \pi_d/2) + (1 - \alpha)/4.$$

Fixing Naive PageRanks

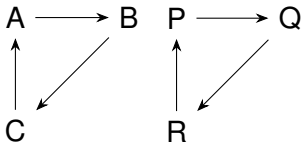
- Why adding a damping factor helps against disconnected graph?
- Start at a random node and follow a random outgoing link at each step. The pagerank of a node v is the probability that you are at v at any given step.



For example, in this graph, at any step, we are at A, B, C with probability $1/3$ (which are their pageranks).

Fixing Naive PageRanks

- But the “probabilities” are not well-defined in this graph because it depends on where you start. You will either get $(1/3, 1/3, 1/3, 0, 0, 0)$ or $(0, 0, 0, 1/3, 1/3, 1/3)$.



Fixing Naive PageRanks

$$\pi_v = \underbrace{\alpha \left(\sum_{u \rightarrow v \in E} \frac{\pi_u}{d_u} \right)}_{\text{with probability } \alpha, \text{ follows an outgoing link}} + \underbrace{\frac{1 - \alpha}{n}}_{\text{with probability } 1 - \alpha \text{ go to a random node}}$$

This makes the graph connected and the distribution exists.

A Brief Review of Matrix Multiplication

Example: Multiply a 2×3 matrix by a 3×2 matrix.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{pmatrix}.$$

$$C = AB = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}, \quad \begin{aligned} c_{11} &= 1 \cdot 7 + 2 \cdot 9 + 3 \cdot 11 = 58, \\ c_{12} &= 1 \cdot 8 + 2 \cdot 10 + 3 \cdot 12 = 64, \\ c_{21} &= 4 \cdot 7 + 5 \cdot 9 + 6 \cdot 11 = 139, \\ c_{22} &= 4 \cdot 8 + 5 \cdot 10 + 6 \cdot 12 = 154. \end{aligned}$$

Hence,

$$C = \begin{pmatrix} 58 & 64 \\ 139 & 154 \end{pmatrix}.$$

A Brief Review of Matrix Multiplication

- Let A be an $m \times n$ matrix and B be an $n \times p$ matrix.

$$A = [a_{ik}]_{1 \leq i \leq m, 1 \leq k \leq n}, \quad B = [b_{kj}]_{1 \leq k \leq n, 1 \leq j \leq p}.$$

A Brief Review of Matrix Multiplication

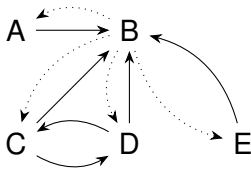
- The entry c_{ij} of the product $C = AB$ is given by

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}, \quad 1 \leq i \leq m, 1 \leq j \leq p.$$

- Equivalently, each $c_{ij} = a_{i*} \cdot b_{*j}$

$$(\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B).$$

Solving system of linear equations with Julia



$$\begin{bmatrix} \pi_A \\ \pi_B \\ \pi_C \\ \pi_D \\ \pi_E \end{bmatrix} = 0.85 \cdot \begin{bmatrix} 0 & \frac{1}{4} & 0 & 0 & 0 \\ 1 & 0 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & \frac{1}{4} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \pi_A \\ \pi_B \\ \pi_C \\ \pi_D \\ \pi_E \end{bmatrix} + 0.15 \cdot \begin{bmatrix} \frac{1}{5} \\ \frac{1}{5} \\ \frac{1}{5} \\ \frac{1}{5} \\ \frac{1}{5} \end{bmatrix}$$

$$\pi = \alpha M \pi + (1 - \alpha) \frac{1}{n} \mathbf{1} \iff (I - \alpha M) \pi = (1 - \alpha) \frac{1}{n} \mathbf{1}$$

PageRank Calculation in Julia

```
using LinearAlgebra

 $\alpha$  = 0.85
n = 5

M = [
    0.0  1/4  0.0  0.0  0.0
    1.0  0.0  1/2  1/2  1.0
    0.0  1/4  0.0  1/2  0.0
    0.0  1/4  1/2  0.0  0.0
    0.0  1/4  0.0  0.0  0.0
]

b = (1 -  $\alpha$ ) / n .* ones(n)

# Solve (I -  $\alpha$ M) $\pi$  = b
 $\pi$  = (I -  $\alpha$ *M) \ b

id_to_label = Dict{1 => "a", 2 => "b", 3 => "c", 4 => "d", 5 => "e"}

for i in 1:length( $\pi$ )
    println(" $\pi$ _", id_to_label[i], " = ",  $\pi$ [i])
end

println()
println("sum( $\pi$ ) = ", sum( $\pi$ ))
```

PageRank Calculation in Julia

$\pi_a = 0.112054244098443$

$\pi_b = 0.3861376192867905$

$\pi_c = 0.19487694625816168$

$\pi_d = 0.19487694625816168$

$\pi_e = 0.112054244098443$

$\text{sum}(\pi) = 0.9999999999999999$

The Iterative Method

Computing Pageranks

- We showed how to compute pageranks by solving a system of linear equations.
- In general, solving a system of n linear equations takes $O(n^3)$ time. Even for $n = 10^6$, this would be infeasible.

Iterative method

- We compute it iteratively (the power method).
- Start with $\pi^{(0)} = (1/n, 1/n, \dots, 1/n)$.
- For $t = 1, 2, \dots$:

$$\pi_v^{(t)} = \alpha \sum_{u \rightarrow v \in E} \frac{\pi_u^{(t-1)}}{d_u} + (1 - \alpha) \frac{1}{n}.$$

- $\pi^{(t)}$ converges to the true pageranks π .
- The running time is $O(m + n)$ per iteration where m is the number of edges.

Computing PageRank

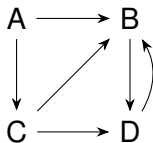
$$d = \frac{(1 - \alpha)}{n}$$

$$\pi_A^{(t+1)} = d$$

$$\pi_B^{(t+1)} = d + \alpha \left(\frac{\pi_A^{(t)}}{2} + \frac{\pi_C^{(t)}}{2} + \pi_D^{(t)} \right)$$

$$\pi_C^{(t+1)} = d + \alpha \left(\frac{\pi_A^{(t)}}{2} \right)$$

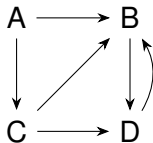
$$\pi_D^{(t+1)} = d + \alpha \left(\pi_B^{(t)} + \frac{\pi_C^{(t)}}{2} \right)$$



Matrix notation

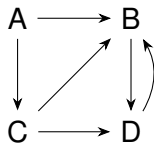
$$\pi^{(t+1)} = \alpha M \pi^{(t)} + \frac{1-\alpha}{n} \mathbf{1}$$

$$M = \begin{bmatrix} 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 1 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \end{bmatrix}$$



A few iterations

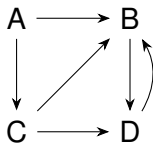
$$\pi^1 = 0.85 \begin{bmatrix} 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 1 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} + 0.15 \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$



$$\pi^1 = \begin{bmatrix} \pi_A^{(1)} \\ \pi_B^{(1)} \\ \pi_C^{(1)} \\ \pi_D^{(1)} \end{bmatrix} = \begin{bmatrix} 0.0375 \\ 0.4625 \\ 0.14375 \\ 0.35625 \end{bmatrix}$$

A few iterations

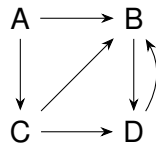
$$\pi^2 = 0.85 \begin{bmatrix} 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 1 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 0.0375 \\ 0.4625 \\ 0.14375 \\ 0.35625 \end{bmatrix} + 0.15 \begin{bmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \end{bmatrix}$$



$$\pi^2 = \begin{bmatrix} 0.0375 \\ 0.41734375 \\ 0.0534375 \\ 0.49171875 \end{bmatrix}$$

A few iterations

$$\pi^3 = 0.85 \begin{bmatrix} 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 1 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 0.0375 \\ 0.41734375 \\ 0.0534375 \\ 0.49171875 \end{bmatrix} + 0.15 \begin{bmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \end{bmatrix}$$



$$\pi^3 = \begin{bmatrix} 0.0375 \\ 0.49410937 \\ 0.0534375 \\ 0.41495312 \end{bmatrix}$$

A few iterations

$$\pi^{10} = \begin{bmatrix} \pi_A^{(10)} \\ \pi_B^{(10)} \\ \pi_C^{(10)} \\ \pi_D^{(10)} \end{bmatrix} = \begin{bmatrix} 0.0375 \\ 0.44753171 \\ 0.0534375 \\ 0.46153079 \end{bmatrix}$$

$$\pi^{11} = \begin{bmatrix} \pi_A^{(11)} \\ \pi_B^{(11)} \\ \pi_C^{(11)} \\ \pi_D^{(11)} \end{bmatrix} = \begin{bmatrix} 0.0375 \\ 0.46844961 \\ 0.0534375 \\ 0.44061289 \end{bmatrix}$$

Convergence

$$\text{error}(t) = \sum_v |\pi_v^{(t)} - \pi_v|.$$

One can show that

Theorem

$$\text{error}(t) = \alpha \sum_v |\pi_v^{(t-1)} - \pi_v|.$$

For $\alpha = 0.85$, $\text{error}(10) = 0.85^{20} \cdot \text{error}(0) \leq 0.038 \cdot \text{error}(0)$.
Error goes down exponentially in terms of the number of iterations.